Disease Spectrum-aware and Time-evolving Dependency Learning for Medication Recommendation

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Medication recommendation is a core component of clinical decision support, aiming to tailor effective drug combinations for patients based on longitudinal Electronic Health Records (EHRs). Despite the success of deep learning in this domain, existing methods typically suffer from two fundamental limitations: (1) they treat diagnosis codes as orthogonal labels, thereby fragmenting shared pharmacological treatment patterns among related diagnoses within latent therapeutic disease spectra, and (2) they infer disease evolution solely based on a coarse-grained holistic hidden state, thereby obscuring the fine-grained temporal dependencies across visit sequences. To address these gaps, we propose SpecTD-MR, a novel medication

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recommendation framework for Disease Spectrum-aware and Time-evolving Dependency Learning. Specifically, we design a Disease Spectrum-aware Hypergraph Learning module that constructs a hypergraph initialized with multi-source clinical knowledge. By employing task-guided clustering with contrastive alignment, this module unifies disparate diagnoses into cohesive, spectrum-aware representations. Furthermore, we introduce a Time-evolving Dependency Modeling module that explicitly quantifies the dependency strength of current diseases on historical contexts. By incorporating a Mixture-of-Experts (MoE) mechanism, this module synergizes disease spectrum with time-evolving dependency to adaptively regulate patient health state transitions. Extensive experiments on real-world datasets demonstrate that SpecTD-MR surpasses state-of-the-art baselines in accuracy, while providing visualizable structural associations to support insights into dynamic disease evolution. [1]

CCS Concepts: • **Social and professional topics → Medical records**; • **Information systems → Recommender systems**.

Additional Key Words and Phrases: Medication Recommendation, Disease Evolution Modeling

1 Introduction

Medication recommendation is a pivotal component of clinical decision support systems, designed to assist clinicians in tailoring effective drug combinations for individual patients [1, 29]. With the widespread adoption of electronic health records (EHRs), the availability of large-scale longitudinal data encompassing diagnoses, procedures, and medications offers an unprecedented opportunity to learn data-driven prescribing patterns [38, 40, 52]. Importantly, the goal of such systems is not to replace clinical expertise, but to support decision-making in increasingly complex and time-constrained healthcare environments, where clinicians must synthesize heterogeneous and longitudinal patient information under significant cognitive constraints [6, 11]. In contemporary clinical practice, physicians are often required to manage patients with multiple comorbidities, long-term disease trajectories, and frequently evolving medication regimens. This requires continuous integration of longitudinal patient history, disease progression patterns, and prior treatment responses, while simultaneously adhering to population-level clinical guidelines and safety constraints. Such integration is difficult to perform reliably through manual reasoning alone [8, 26]. As a result, medication recommendation systems have the potential to reduce adverse drug events, alleviate physician workload, and promote more consistent, evidence-based treatment decisions, especially for patients with complex and evolving clinical profiles [21, 50]. Developing medication recommendation models that can effectively organize and reason over the temporal, relational, and evolving structure of longitudinal EHR data is therefore a problem of practical significance for healthcare information systems.

In recent years, deep learning techniques have demonstrated significant potential in modeling longitudinal EHR data for medication recommendation. Early approaches primarily relied on Recurrent Neural Networks (RNNs) to capture temporal dependencies in patient history [7, 13, 50]. More recently, methods incorporating Graph Neural Networks (GNNs) and Memory Networks [2, 14, 19, 24, 33, 37, 39, 44] have advanced the field by modeling Drug-Drug Interactions (DDI) [32, 43, 48] knowledge or enhancing patient longitudinal representation learning. These methods typically embed medical concepts into low-dimensional vector spaces to predict medication sets. Despite their success, existing approaches often oversimplify the intricate latent disease spectra among diagnoses and the time-evolving dependencies of disease progression. Such oversimplification essentially leads to two challenges critical for reliable medication recommendations, but unaddressed in the literature: (1) how to effectively capture the broader disease spectra that group clinically related diagnoses sharing similar pharmacological treatments; and (2) how to explicitly model the fine-grained time-evolving dependencies between current and historical diseases along a patient's trajectory.

First, diagnosis codes may exhibit broader clinical and therapeutic regularities, where related conditions share pharmacological treatment patterns even when they do not strictly belong to the same anatomical or etiological taxonomy. In this work, we refer to such task-relevant, treatment-oriented groupings as latent therapeutic disease

[1] Our code is available at: https://github.com/liyifo/SpecTD-MR.

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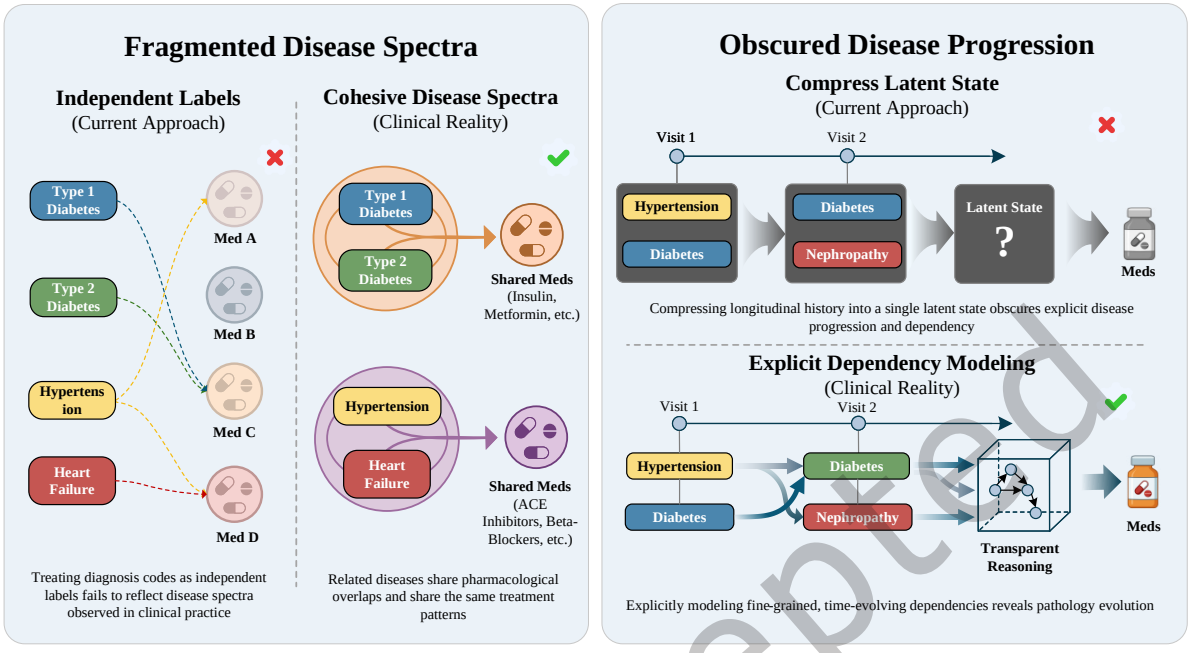


Fig. 1. Examples illustrating limitations of existing EHR-based medication recommendation abstractions. (a) Treating diagnosis codes as independent labels fails to reflect disease spectra observed in clinical practice. (b) Compressing longitudinal history into a single latent state obscures explicit time-evolving dependency of disease progression.

spectra [34, 45, 46]. However, existing methods typically treat codes as orthogonal labels [18, 19, 36], a formulation that artificially severs the semantic connections within these latent disease spectra. For instance, although *Type 1* and *Type 2 diabetes* share significant pharmacological overlaps, treating them as distinct, isolated categories forces the model to learn their prescribing behaviors separately. This fragmentation dilutes the supervision signals for each code and prevents the model from leveraging cross-code regularities. Hence, a therapeutic disease spectrum-aware representation that can jointly encode clusters of related diseases is crucial to better capture shared treatment patterns and improve recommendation performance.

Second, a patient's EHR reflects a dynamic process where current diseases often evolve temporally from previous conditions [4, 15, 22, 31]. However, existing sequence-based methods [5, 22, 39, 43] typically infer disease evolution solely based on a coarse-grained holistic hidden state. This uninterpretable "black-box" inference mechanism blurs fine-grained disease evolutionary pathways into a generic context, thereby making it difficult to distinguish between incidental comorbidity co-occurrences and meaningful temporal progressions. For instance, a diagnosis of *nephropathy* is not merely a new code, but often a downstream complication predictively linked to prior *hypertension* and *diabetes*. Implicit models lack this structural awareness and tend to obscure such critical predictive dependency signals. Consequently, they often miss the specific triggers for medication adjustment and fail to provide the interpretable insights essential for trusted clinical decision support.

To tackle the above challenges, we propose a novel Disease **Spec**trum-aware and **T**ime-evolving **D**ependency-based **M**edication **R**ecommendation (**SpecTD-MR**) framework. Specifically, to address the first challenge, we design a **Disease Spectrum-aware Hypergraph Learning module**. This module constructs a hypergraph based

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on various patients and initializes the nodes by integrating structural, textual, and logical clinical knowledge, thereby injecting explicit semantic relationships between clinically related codes into the graph topology. To overcome the fragmentation of disease spectra, we employ task-guided clustering with alignment. This mechanism forces distinct concepts that share pharmacological treatment patterns to cluster together in the latent space, effectively unifying disparate concept nodes and visit hyperedges into cohesive, spectrum-aware representations that share statistical strength across the entire spectrum. To address the second challenge, we introduce a **Time-evolving Dependency Modeling module**. By explicitly modeling the temporal relevance between current diseases with the conditions of historical visits, this module constructs a structured evolution matrix to capture the fine-grained trajectory of patient health states. Furthermore, the module incorporates a Mixture-of-Experts (MoE) mechanism to leverage latent disease spectrum information. This design synergizes latent disease spectrum with time-evolving dependency to adaptively regulate the transition of patient health states.

In summary, the main contributions of this paper are as follows:

- We propose SpecTD-MR, a unified framework that synergistically addresses the limitations of fragmented disease spectra and coarse-grained implicit disease evolution. By rethinking diagnosis codes as interconnected components of latent disease spectra and shifting from holistic hidden states to explicit dependency inference, SpecTD-MR offers a more clinically motivated approach to EHR modeling.
- We develop a Disease Spectrum-aware Hypergraph Learning module to overcome the fragmentation of disease spectra. Unlike existing methods that isolate clinically related diagnoses, we employ task-guided dual-path clustering with contrastive alignment. This novel approach unifies disparate concepts with underlying pharmacological correlations into cohesive representations, effectively bridging the semantic gap in sparse clinical data.
- We introduce a Time-evolving Dependency Modeling module to explicitly model the predictive relevance between adjacent visits and construct a fine-grained evolution matrix. This module enables adaptive state transitions based on disease evolution intensity rather than generic temporal updates.
- Extensive experiments on MIMIC-III and MIMIC-IV demonstrate that SpecTD-MR clearly outperforms state-of-the-art baselines. Beyond accuracy, our framework provides interpretable insights for its recommendations by visualizing dynamic disease dependencies and spectrum structures, which is crucial for trusted clinical decision support.

2 Related Work

2.1 Medication Recommendation from EHRs

Medication recommendation from EHRs can be broadly divided into *instance-based* and *longitudinal* methods. Instance-based methods [13, 50] rely only on the current visit, encoding present diagnoses and procedures to predict medications. For example, LEAP [50] uses attention over instances within the current visit under a multi-instance multi-label framework. However, ignoring historical visits makes it difficult to capture long-term disease progression and recurrent treatment patterns.

Longitudinal methods further exploit historical EHRs together with the current visit to model patient trajectories [2, 5, 14, 22, 43, 44, 47]. COGNet [39] proposes a copy-or-predict mechanism that reuses historical medications. ARMR [36] utilizes segmented temporal learning components to extract changes in patient status, and MR-DTR [22] encodes longitudinal history access by considering relative positions in the temporal domain, learning time-sensitive patient representations. These models significantly improve performance over instance-based baselines.

However, they typically treat each diagnosis code as an isolated category and mostly rely on code or visit-level co-occurrence. As a result, they fail to explicitly organize related diagnoses into disease spectra with shared

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therapeutic patterns, and they do not carefully model time-evolving dependencies between current and past diseases.

2.2 Graph-based Representation Learning for Medication Recommendation

GNNs have been widely used to obtain structure-aware representations for medication recommendation. A large body of work builds GNNs on molecular graphs to encode drug structures. SafeDrug [43] designs a dual molecular graph encoder to capture both global and local molecular features. MoleRec [44] focuses on molecular substructures and their impact on patient health, and GPSRec [14] combines pre-training methods with dual-threshold loss adjustment, effectively mitigating the risk of undesirable side effects. These methods demonstrate the effectiveness of graph-based molecular modeling, but they operate primarily at the drug level and do not explicitly organize diseases into higher-level spectra or subtypes.

Another branch of work uses GNNs to model relations among clinical concepts. GAMENet [30] and REFINE [3] construct EHR and DDI graphs to capture interactions among drugs and between drugs and diagnoses. However, most of these models are built on pairwise graphs where each diagnosis code is treated as a separate entity. They rarely exploit hypergraph structures or explicit clustering mechanisms that can connect multiple diagnoses and visits within the same disease spectrum. As a result, they struggle to learn disease-spectrum-aware representations that reveal shared treatment patterns and subtype structures.

2.3 Temporal Modeling of Disease Progression in Medication Recommendation

Because EHRs are inherently sequential, many models incorporate temporal architectures to capture disease progression. RETAIN [7] uses reverse-time attention to highlight influential past visits, and memory-augmented models such as GAMENet [30] store and retrieve historical information to assist current predictions. PREMIER [2] and related methods further exploit visit order and, to some extent, time intervals. However, most existing methods aggregate historical visits into a small number of hidden states or context vectors, without explicitly modeling how dependencies between current and past diseases evolve over time. These longitudinal models confirm that historical information is critical for accurate recommendation.

In particular, MR-DTR [22] initially focused on the dynamic evolution of diseases, but its reliance on prescriptions from similar patients still fails to model disease dependency at a granular level. These methods focus on visit-level patterns rather than constructing time-evolving dependency structures among individual diseases, and seldom adapt their transition dynamics according to changes in disease–disease relations. This makes it difficult to distinguish different stages of disease development and to align medication strategies with temporal evolution patterns, which motivates our spectrum-aware hypergraph representation learning and time-evolving dependency modeling.

3 Preliminary

Electronic Health Records (EHR). Let $\mathcal{C} = \mathcal{D} \cup \mathcal{P} \cup \mathcal{M}$ denote the comprehensive set of medical concepts, where $\mathcal{D}$, $\mathcal{P}$, and $\mathcal{M}$ represent the sets of diagnoses, procedures, and medications, respectively. The total number of unique concepts is denoted as $N = |\mathcal{C}|$. The EHR of a patient u is represented as a longitudinal sequence of visits $\mathcal{V}^u = [\mathbf{v}_1^u, \mathbf{v}_2^u, \ldots, \mathbf{v}_t^u]$. Each visit $\mathbf{v}^u$ is characterized by a subset of concepts $\mathcal{C}_t \subseteq \mathcal{C}$ (including diagnoses, procedures, and medications), with $N_t = |\mathcal{C}_t|$. $\mathbf{v}_t^u$ excludes medications because the medications constitute the prediction target $\mathbf{y}_t \in \{0,1\}^{|\mathcal{M}|}$. Additionally, let $\mathbf{A}_t^{prior} \in \mathbb{R}^{N_t \times N_t}$ denote the prior co-occurrence matrix for concepts in $\mathcal{C}_t$, derived from whole population.

Hypergraph-based abstraction. We construct a concept-level hypergraph $\mathcal{H} = (\mathcal{C}, \mathcal{E})$ from the EHRs. Here, nodes correspond to medical concepts (diagnoses, procedures and medications), and each hyperedge $e \in \mathcal{E}$ represents a patient visit, connecting all concepts co-occurring in that visit. Medications that are the prediction

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Table 1. Key notations and descriptions.

Symbol	Descriptions
$\mathcal{D}, \mathcal{P}, \mathcal{M}$	Diagnosis, procedure, and medication vocabularies.
$\mathcal{C}$, N	Global medical concept set $\mathcal{C} = \mathcal{D} \cup \mathcal{P} \cup \mathcal{M}$; $N = \lvert\mathcal{C}\rvert$.
$\mathcal{V}^u$	Visit sequence of patient u: $[\mathbf{v}_1^u, \ldots, \mathbf{v}_t^u]$.
$\mathcal{C}_t$, N_t	Active concepts in visit t and its size; $N_t = \lvert\mathcal{C}_t\rvert$.
$\mathbf{y}_t$, $\hat{\mathbf{y}}_t$	Ground-truth multi-hot medications and predicted medication probabilities (last visit).
$\mathbf{A}_t^{prior}$	Prior co-occurrence matrix for concepts in visit t.
$\mathcal{H} = (\mathcal{C}, \mathcal{E})$	Concept-level hypergraph; each $e \in \mathcal{E}$ corresponds to a visit.
$\mathbf{I}$	Incidence matrix of the hypergraph, $\mathbf{I} \in \{0,1\}^{N \times \lvert\mathcal{E}\rvert}$.
d	Latent embedding dimension.
$\mathbf{H}_{\mathcal{C}}, \mathbf{H}_{\mathcal{E}}$	Hypergraph encoder outputs: concept embeddings and visit embeddings.
K	Number of clusters (and prototypes if shared).
$\mathbf{H}_t$	Visit input matrix (special token + active concept embeddings).
$\tilde{\mathbf{Q}}_t, \tilde{\mathbf{K}}_t, \mathbf{V}_t$	Current-view query/key/value projections of $\mathbf{H}_t$.
$\mathbf{Q}_t, \mathbf{K}_t$	History-aware query/key after fusion with cached states (union-aligned).
$\mathbf{S}_t$	Time-evolving dependency matrix for visit t.
$\mathbf{g}_t$	Visit-level readout state extracted from the special token.
$_t^{adj}$, $\mathbf{v}_{prog,t}$	Dependency variation $\lvert\mathbf{S}_t - \mathbf{S}_{t-1}\rvert$ and progress feature.
, α_t	Prototype bank and routing weights in the MoE mechanism.
$\mathbf{h}_t$, γ_t	PDD-GRU hidden state and evolution gate.

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targets for the last visit are excluded from hypergraph nodes to prevent label leakage. Historical medications (from earlier visits) may be included as contextual nodes. We denote the incidence matrix as $\mathbf{I} \in \{0,1\}^{N \times |\mathcal{E}|}$, where $\mathbf{I}_{i,j} = 1$ if concept node c_i is contained in visit hyperedge e_j. The hypergraph encoder produces the final concept-level embeddings $\mathbf{H}_{\mathcal{C}} \in \mathbb{R}^{N \times d}$ and visit-level embeddings $\mathbf{H}_{\mathcal{E}} \in \mathbb{R}^{|\mathcal{E}| \times d}$, where d is the latent dimension.

Medication recommendation. Given the EHR sequence $\mathcal{V}^u$ of a patient, the goal of medication recommendation is to predict the optimal medication combination $\hat{\mathbf{y}}_t \in \{0,1\}^{|\mathcal{M}|}$ for the last visit. The key notations in this paper are listed in Table 1.

4 Methodology

In this section, we present **SpecTD-MR**, our framework for medication recommendation. As illustrated in Figure 2, the model integrates two mutually reinforcing modules. First, the *Disease Spectrum-aware Hypergraph Learning* module constructs a hypergraph from multi-source semantic features and captures high-order correlations, uncovering latent therapeutic disease spectra via task-guided dual-path clustering and contrastive alignment. Second, the *Time-evolving Dependency Modeling* module explicitly quantifies the dependency strength by fusing historical context and models temporal evolution patterns with a Prototype-driven Disease Dependency GRU (PDD-GRU). These modules enable SpecTD-MR to capture both spectrum-aware representations and fine-grained time-evolution dependencies, ensuring clinically meaningful recommendations.

Algorithm 1 SpecTD-MR Training and Inference Pipeline

Require: Patient visit sequences $\{\mathcal{V}^u\}$; hypergraph $\mathcal{H} = (\mathcal{C}, \mathcal{E})$ with incidence matrix $\mathbf{I}$; priors $\{\mathbf{A}_\tau^{prior}\}$.
Ensure: Trained parameters of SpecTD-MR.

Stage I: Hypergraph pretraining (Disease spectrum-aware learning)

Initialize concept features from multi-source semantics $\mathbf{X}_s, \mathbf{X}_{txt}, \mathbf{X}_{log}$.
for each warm-up epoch **do**
  Encode hypergraph and obtain $\mathbf{H}_\mathcal{C}, \mathbf{H}_\mathcal{E}$ via iterative message passing.
  Sample masked concepts $\mathcal{C}_{mask}$ and compute $\mathcal{L}_{mask}$.
  Update the hypergraph encoder by minimizing $\mathcal{L}_{mask}$.
end for
for each pretraining epoch **do**
  Encode hypergraph and obtain $\mathbf{H}_\mathcal{C}, \mathbf{H}_\mathcal{E}$ via iterative message passing.
  Compute the task-guided dual-path clustering and contrastive alignment objective $\mathcal{L}_{pretrain} = \mathcal{L}_{med} + \gamma_1 \mathcal{L}_{DEC} + \gamma_2 \mathcal{L}_{align}$.
  Update the hypergraph encoder by minimizing $\mathcal{L}_{pretrain}$.
end for
Initialize prototype bank using visit-cluster centroids from $\mathbf{H}_\mathcal{C}$.

Stage II: Downstream training (Time-evolving dependencies + recommendation)

for each training epoch **do**
  for each patient u with visits $\mathcal{V}^u = [\mathbf{v}_1^u, \ldots, \mathbf{v}_t^u]$ **do**
    Initialize cached states $\mathbf{Q}_0 = \mathbf{K}_0 = \mathbf{0}$ (union-aligned cache).
    for $\tau = 1$ to t **do**
      Form $\mathbf{H}_\tau$ (special token + active concepts) and compute $\tilde{\mathbf{Q}}_\tau, \tilde{\mathbf{K}}_\tau, \mathbf{V}_\tau$.
      Fuse cached states to obtain $\mathbf{Q}_\tau, \mathbf{K}_\tau$ and infer dependencies $\mathbf{S}_\tau$ using $\mathbf{A}_\tau^{prior}$.
      Update node features and readout $\mathbf{g}_\tau$ from the special token.
      Compute τ^{adj} and $\mathbf{v}_{prog,\tau}$; route prototypes α_τ; update PDD-GRU state $\mathbf{h}_\tau$ with evolution gate γ_τ.
    end for
    Predict $\hat{\mathbf{y}}_t = \sigma(\mathbf{W}_{out}[\mathbf{h}_t \parallel \mathbf{g}_t] + \mathbf{b}_{out})$ on the last visit.
    Compute $\mathcal{L}_{med}$ using $\mathbf{y}_t$ and update all parameters by backpropagation.
  end for
end for

Inference: For patient u, output $\hat{\mathbf{y}}_t$ and recommend medication i if $\hat{\mathbf{y}}_{t,i} > 0.5$.

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4.1 Disease Spectrum-aware Hypergraph Learning

To encompass a broader spectrum of clinically relevant diagnoses that share similar pharmacological treatments, we design a *Disease Spectrum-aware Hypergraph Learning* module. Inspired by TACCO [51], this module first initializes concept representations using multi-source knowledge and learns preliminary node associations through a masked prediction task. Subsequently, it optimizes the clustering of the disease spectrum through task-guided dual-path clustering and contrastive alignment.

It is important to clarify that in the context of this medication recommendation framework, the term "disease spectrum" is operationalized as a *task-driven therapeutic spectrum* rather than a strict pathophysiological taxonomy. While established taxonomies (like ICD) group diseases by anatomy or etiology, our goal is to construct cohesive representations for diseases that share underlying *pharmacological treatment patterns*. To ensure these learned clusters do not degenerate into spurious statistical co-occurrences (e.g., incidental age-related comorbidities), the clustering is explicitly guided by two constraints: (1) multi-source semantic initialization, where the logical hierarchy features $\mathbf{X}_{log}$ provide a rigid ontological anchor, while the textual features $\mathbf{X}_{txt}$ capture nuanced clinical descriptions that supplement the limitations of fixed code structures; and (2) task-guided joint optimization, where the clustering is directly supervised by the medication prediction objective $\mathcal{L}_{med}$.

4.1.1 Multi-source Semantic Initialization. To enhance the semantic capability of the module, we utilize three sources of raw features for each medical concept node $i \in \mathcal{C}$: (1) structural features $\mathbf{X}_s$ obtained by performing random walks [27] on the co-occurrence graph; (2) textual features $\mathbf{X}_{txt}$ encoded by SapBERT [23] from descriptions; and (3) logical hierarchy embeddings $\mathbf{X}_{log}$ obtained from Poincaré Embeddings [25] based on ICD/ATC[2] codes. These features are fused via a learnable gating mechanism to form the initial node embedding $\mathbf{H}_{\mathcal{C}}^{(0)} \in \mathbb{R}^d$:

$$\mathbf{H}_{\mathcal{C}}^{(0)} = \text{LayerNorm}(\mathbf{W}_s\mathbf{X}_s \parallel (\alpha \cdot \mathbf{W}_t\mathbf{X}_{txt} + (1-\alpha) \cdot \mathbf{W}_l\mathbf{X}_{log})), \tag{1}$$

where $\parallel$ denotes concatenation, $\mathbf{W}_*$ are projection matrices mapping inputs to dimension d, and the gating coefficient α is obtained as $\alpha = \sigma(\text{FFN}[X_{\text{txt}} \| X_{\text{log}}])$.

4.1.2 Hypergraph Encoding. We adopt a Transformer block to propagate information. The message passing process updates concept node representations $\mathbf{H}_{\mathcal{C}}$ and visit hyperedge representations $\mathbf{H}_{\mathcal{E}}$ iteratively. For the l-th layer:

$$\mathbf{H}_{\mathcal{E}}^{(l+1)} = \rho_e\left(\mathbf{I}^\top \mathbf{H}_{\mathcal{C}}^{(l)}\right), \quad \mathbf{H}_{\mathcal{C}}^{(l)} = \rho_c\left(\mathbf{I}\,\mathbf{H}_{\mathcal{E}}^{(l)}\right), \tag{2}$$

where $\rho_*(\mathbf{X}) = \text{LayerNorm}(\mathbf{Z} + \text{FFN}(\mathbf{Z})), \mathbf{Z} = \text{LayerNorm}(\mathbf{X} + \text{MHSA}(\mathbf{X}))$, $\rho_*(\cdot)$ is a transition function parameterized by a Multi-Head Self-Attention (MHSA) block followed by a Feed-Forward Network (FFN), similar to Transformer layers. This iterative process effectively captures high-order correlations between medical concepts and visit patterns.

4.1.3 Masked Concept Prediction Warm-up. Before performing spectrum-level clustering, we first warm up the hypergraph encoder via a self-supervised masked concept prediction task. For each visit hyperedge, we randomly mask a subset of concept nodes and train the encoder to recover their identities from the remaining concepts and the hypergraph context. Formally, let $\mathcal{C}_{\text{mask}}$ denote the masked concepts within a visit, the objective is the negative log-likelihood of the masked concepts:

$$\mathcal{L}_{mask} = -\sum_{c \in \mathcal{C}_{\text{mask}}} \log(\text{Softmax}\big(\text{FFN}(\mathbf{h}_c)\big)[c]), \tag{3}$$

where $\mathbf{h}_c$ is the hypergraph-encoded representation of the masked node. This warm-up encourages the encoder to capture intrinsic co-occurrence patterns and latent spectrum-level relationships among concepts, providing a strong initialization for subsequent clustering.

4.1.4 Task-guided Dual-path Clustering and Alignment. To uncover latent disease spectra, we perform Deep Embedded Clustering (DEC) [41] on both the concept space and the visit space. Let $\mathbf{h}_i^{\mathcal{C}}$ and $\mathbf{h}_j^{\mathcal{E}}$ denote the final embeddings for concept i and visit j, respectively. We maintain learnable cluster centroids $\{\mu_k^{\mathcal{C}}\}_{k=1}^K$ for concepts and $\{\mu_k^{\mathcal{E}}\}_{k=1}^K$ for visits. We compute the soft assignment probability q_{ik} of node/visit i to cluster k using the Student's t-distribution, and define a target distribution p_{ik} that sharpens high-confidence assignments. The clustering loss is the KL divergence:

$$\mathcal{L}_{DEC} = \text{KL}(\mathbf{P}_{\mathcal{V}} \| \mathbf{Q}_{\mathcal{V}}) + \text{KL}(\mathbf{P}_{\mathcal{E}} \| \mathbf{Q}_{\mathcal{E}}), \tag{4}$$

where $q_{ik} = \frac{(1+\|\mathbf{h}_i^* - \mu_k^*\|^2)^{-1}}{\sum_j (1+\|\mathbf{h}_i^* - \mu_j^*\|^2)^{-1}}$ and $p_{ik} = \frac{q_{ik}^2 / \sum_i q_{ik}}{\sum_j (q_{jk}^2 / \sum_i q_{jk})}$.

[2]https://www.whocc.no/atc/structure_and_principles/

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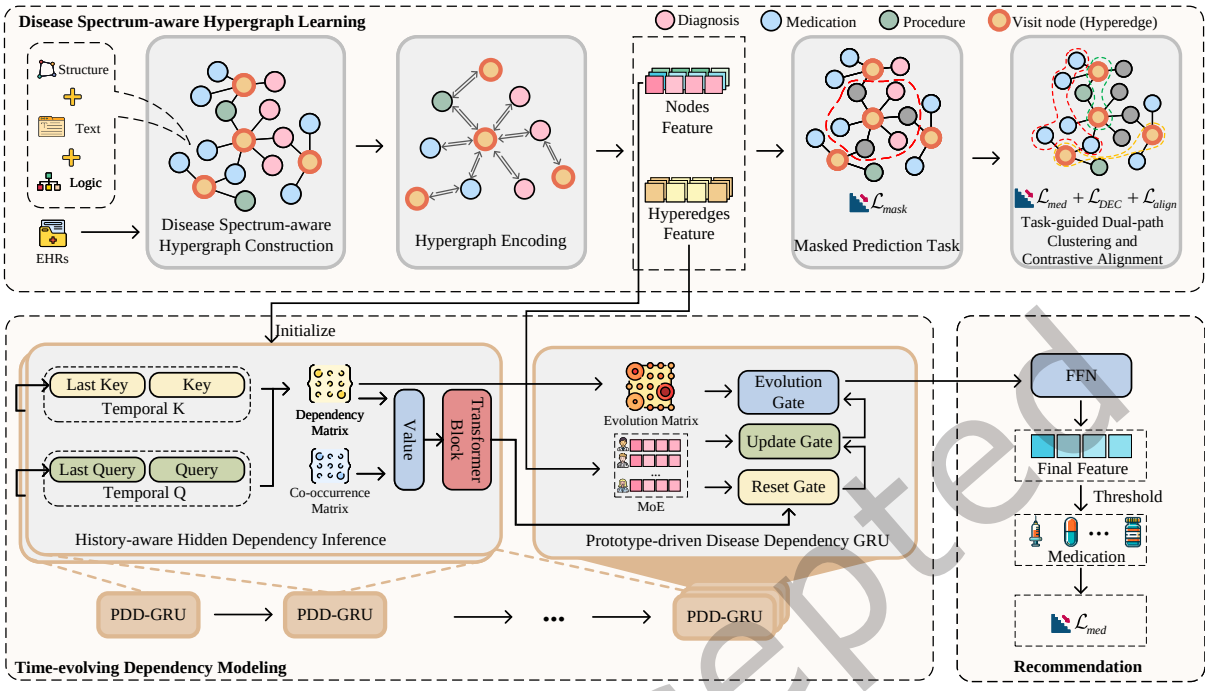


Fig. 2. The overall architecture of the proposed SpecTD-MR framework. It comprises two key modules: a Disease Spectrum-aware Hypergraph Learning module and a Time-evolving Dependency Modeling module. The former initializes concept representations from multi-source semantics and uncovers latent disease spectra via masked prediction and task-guided clustering and alignment. The latter infers dynamic visit dependencies and employs a Prototype-driven Disease Dependency GRU to model the disease evolution of health states.

Then, to couple the concept space with the trajectory space, we compute cluster-level centroids:

$$\mathbf{c}_k^{\mathcal{V}} = \frac{\sum_{i=1}^{N} q_{ik}^{\mathcal{V}} \mathbf{z}_i^{\mathcal{V}}}{\sum_{i=1}^{N} q_{ik}^{\mathcal{V}}}, \quad \mathbf{c}_k^{\mathcal{E}} = \frac{\sum_{j=1}^{M} q_{jk}^{\mathcal{E}} \mathbf{z}_j^{\mathcal{E}}}{\sum_{j=1}^{M} q_{jk}^{\mathcal{E}}}. \tag{5}$$

Then, We project the concept and visit clusters using FFN($\cdot$). For each concept cluster $\mathbf{u}_k = \text{FFN}(\mathbf{c}_k^{\mathcal{V}})$, we identify its most similar visit cluster $\mathbf{v}_{pos(k)} = \text{FFN}(\mathbf{c}_{pos(k)}^{\mathcal{E}})$ as the positive pair, while treating all other visit clusters as negative samples $\mathbf{v}_{neg(k)}$. The alignment loss enforces a margin δ between positive and negative pairs:

$$\mathcal{L}_{align} = \frac{1}{K} \sum_{i=1}^{K} \sum_{j \neq pos(i)} \max\left(0, \text{dist}(\mathbf{u}_i, \mathbf{v}_{pos(i)}) - \text{dist}(\mathbf{u}_i, \mathbf{v}_{neg(j)}) + \delta\right), \tag{6}$$

where $\text{dist}(\mathbf{a}, \mathbf{b}) = 1 - \text{CosineSimilarity}(\mathbf{a}, \mathbf{b})$. To align the representations with the downstream goal, we introduced a supervised loss guided by the drug recommendation task. The total pre-training objective is $\mathcal{L}_{pretrain} = \mathcal{L}_{med} + \gamma_1 \mathcal{L}_{DEC} + \gamma_2 \mathcal{L}_{align}$, $\mathcal{L}_{med}$ is consistent with Section 4.4.

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4.2 Time-evolving Dependency Modeling

To explicitly capture the dynamic progression of health states and the disease evolution between visits, we propose the Time-evolving Dependency Modeling module. This module consists of a History-aware Hidden Dependency Inference mechanism to construct time-variant dependency graphs, and a Prototype-driven Disease Dependency GRU to model the evolution trajectory of disease.

4.2.1 History-aware Hidden Dependency Inference. This mechanism constructs a dynamic dependency matrix for the visit by fusing historical context. Let $\mathbf{H}_t \in \mathbb{R}^{N_t \times d}$ denote the visit-level input matrix for the t-th visit. It is constructed from (i) the pre-trained concept embeddings for the active medical codes in the current visit, and (ii) a learnable visit-level special token that serves as a global readout node. Formally, given the global concept embedding matrix $\mathbf{H}_{\mathcal{C}} \in \mathbb{R}^{|\mathcal{C}| \times d}$ from Section 4.1, we first gather the embeddings of active concepts $\mathcal{C}_t$ to form $\mathbf{H}_t^{(c)} \in \mathbb{R}^{|\mathcal{C}_t| \times d}$, and then prepend a special token initialized by the hyperedge (visit) embedding $\mathbf{e}_t \in \mathbb{R}^d$:

$$\mathbf{H}_t = \begin{bmatrix} \mathbf{e}_t \\ \mathbf{H}_t^{(c)} \end{bmatrix} \in \mathbb{R}^{(1+|\mathcal{C}_t|) \times d}, \tag{7}$$

where $N_t = 1 + |\mathcal{C}_t|$ and the first row corresponds to the special token. For unseen visits in the validation or test set, a corresponding hyperedge is constructed, and the pretrained hypergraph encoder is applied to obtain the visit-level representation $\mathbf{e}_t$. We then project $\mathbf{H}_t$ into a current-view query $\tilde{\mathbf{Q}}_t$, key $\tilde{\mathbf{K}}_t$, and value $\mathbf{V}_t$:

$$\tilde{\mathbf{Q}}_t = \mathbf{H}_t \mathbf{W}_q, \quad \tilde{\mathbf{K}}_t = \mathbf{H}_t \mathbf{W}_k, \quad \mathbf{V}_t = \mathbf{H}_t \mathbf{W}_v. \tag{8}$$

To explicitly model the temporal evolution, we carry over the query and key states from the previous visit, denoted as $\mathbf{Q}_{t-1}$ and $\mathbf{K}_{t-1}$. We introduce fusion matrices $\mathbf{W}_{qf}, \mathbf{W}_{kf} \in \mathbb{R}^{2d \times d}$ to integrate the current view with the historical view. The history-aware query $\mathbf{Q}_t$ and key $\mathbf{K}_t$ are computed via concatenation and linear fusion:

$$\mathbf{Q}_t = \mathbf{W}_{qf}[\tilde{\mathbf{Q}}_t \parallel \mathbf{Q}_{t-1}], \quad \mathbf{K}_t = \mathbf{W}_{kf}[\tilde{\mathbf{K}}_t \parallel \mathbf{K}_{t-1}], \tag{9}$$

where we define $\mathbf{Q}_0$ and $\mathbf{K}_0$ as zero tensors. The fused $\mathbf{Q}_t$ and $\mathbf{K}_t$ are used both for the current dependency inference and for updating the historical cache. Note that the active concept set $\mathcal{C}_t$ (and thus N_t) varies across visits. To make the fusion of current and historical states well-defined, we align the query and key representations based on the *union* of concepts. Specifically, historical $\mathbf{Q}_{t-1}$ and $\mathbf{K}_{t-1}$ are gathered from a cache indexed by the union of concepts observed so far, and only the rows corresponding to the current visit's nodes are selected for concatenation with $\tilde{\mathbf{Q}}_t$ and $\tilde{\mathbf{K}}_t$. Newly appeared concepts are treated as zero-initialized states, consistent with $\mathbf{Q}_0$ and $\mathbf{K}_0$.

This design shares a high-level motivation with Transformer-XL [9]'s segment-level recurrence, which reuses historical states to extend temporal context. However, EHR visits consist of variable-sized medical concept sets rather than fixed token sequences. Therefore, we maintain concept-indexed Q/K caches aligned by the union of observed concepts. Unlike directly reusing past K/V states, SpecTD-MR fuses current and historical Q/K states through learnable projections, enabling the model to track changes in predictive association patterns across visits.

We compute the time-evolving dependency matrix $\mathbf{S}_t$ and inject domain priors to guide the learning via the prior co-occurrence matrix $\mathbf{A}_t^{prior}$. The dependency matrix $\mathbf{S}_t \in \mathbb{R}^{N_t \times N_t}$ is calculated as:

$$\mathbf{S}_t = \mathrm{Softmax}\left(\frac{\mathbf{Q}_t \mathbf{K}_t^\top}{\sqrt{d}} + \lambda \log(\mathbf{A}_t^{prior} + \epsilon)\right), \tag{10}$$

where λ controls the strength of the prior bias, ϵ is a small constant for numerical stability. It is important to clarify that the term "dependency" in this work is used in the standard machine-learning sense of predictive/statistical

Just Accepted

dependency. Accordingly, the dependency matrices represent model-inferred predictive dependencies learned from longitudinal EHR data, rather than verified causal, pathological, or mechanistic relationships.

The updated concept features are obtained via attention aggregation followed by a Transformer block:

$$\begin{aligned} \mathbf{O}_t &= \mathbf{S}_t \mathbf{V}_t, \\ \mathbf{Z}_t &= \text{LayerNorm}(\mathbf{H}_t + \mathbf{O}_t \mathbf{W}_o), \\ \mathbf{H}'_t &= \text{LayerNorm}(\mathbf{Z}_t + \text{FFN}(\mathbf{Z}_t)), \end{aligned} \tag{11}$$

where $\mathbf{W}_o \in \mathbb{R}^{d \times d}$ is the output projection matrix. The updated concept features $\mathbf{H}'_t$ now encode both the topological dependencies within the current visit and the evolutionary semantics from the patient's history.

After the dependency inference layers, we extract the updated representation of the special token $\mathbf{h}'_t = \mathbf{H}'_t[0, :]$, and obtain a compact visit-level graph state $\mathbf{g}_t$ via a projection head:

$$\mathbf{g}_t = \text{GELU}\Big(\text{LayerNorm}(\mathbf{W}_{\text{ro}} \mathbf{h}'_t + \mathbf{b}_{\text{ro}})\Big), \tag{12}$$

where $\mathbf{W}_{\text{ro}} \in \mathbb{R}^{d \times d}$ and $\mathbf{b}_{\text{ro}} \in \mathbb{R}^d$ are learnable parameters. The resulting $\mathbf{g}_t$ is then fed into the subsequent PDD-GRU as the structural context for temporal state transitions.

4.2.2 Prototype-driven Disease Dependency GRU. To model temporal evolution patterns, we introduce the PDD-GRU, which generates dynamic evolution parameters based on visit prototypes and regulates state updates via explicit dependency variation awareness.

First, we explicitly measure the dynamic intensity of disease evolution in the patient's condition. We calculate the evolution matrix $\overset{adj}{t}$ as follows:

$$\overset{adj}{t} = |\mathbf{S}_t - \mathbf{S}_{t-1}|. \tag{13}$$

To condense this matrix into a usable feature vector, we first compute the *Degree Delta* $\mathbf{d}_{t,i}$ for each node by summing the absolute changes in its row:

$$\mathbf{d}_{t,i} = \sum_{j=1}^{N_t} \overset{adj}{t}[i, j], \quad i = 1, \ldots, N_t. \tag{14}$$

Then, we aggregate these nodal values by concept type (Diagnosis, Procedure, Medication). This yields the progress feature vector $\mathbf{v}_{prog,t} \in \mathbb{R}^3$:

$$\mathbf{v}^{(c)}_{prog,t} = \begin{cases} \dfrac{1}{|\mathcal{J}^{(c)}_t|} \displaystyle\sum_{i \in \mathcal{J}^{(c)}_t} \mathbf{d}_{t,i}, & |\mathcal{J}^{(c)}_t| > 0, \\ 0, & |\mathcal{J}^{(c)}_t| = 0, \end{cases} \qquad c \in \{\text{diag}, \text{proc}, \text{med}\}, \tag{15}$$

where $\mathcal{J}^{(c)}_t$ denotes the set of active concept indices of type c at visit t. The resulting vector $\mathbf{v}_{prog,t}$ provides a quantitative indicator of how strongly the disease dependency structure has evolved between visits $t-1$ and t.

Then, to capture diverse visit patterns, we employ a Mixture-of-Experts (MoE) mechanism [49]. We maintain a bank of learnable prototypes $\in \mathbb{R}^{K \times d}$. To facilitate robust adaptation, is initialized using the visit cluster centroids of hyperedge obtained from the hypergraph pre-training stage, thereby inheriting the global semantic structures of the disease spectrum. We compute the routing weights $\alpha_t \in \mathbb{R}^K$ via dot-product attention:

$$\alpha_t = \text{Softmax}(\mathbf{g}_t \mathbf{W}_p{}^\top), \tag{16}$$

Just Accepted

where $\mathbf{W}_p \in \mathbb{R}^{d\times d}$ is a projection matrix. Unlike standard GRUs with fixed weights, we maintain K sets of expert weights for the input gates: $\{\mathbf{W}_z^{(k)}, \mathbf{W}_r^{(k)}, \mathbf{W}_h^{(k)}\}_{k=1}^{K}$. The effective weight matrices for the current step are dynamically synthesized:

$$\mathbf{W}_z^{(t)} = \sum_{k=1}^{K} \alpha_{t,k}\mathbf{W}_z^{(k)}, \mathbf{W}_r^{(t)} = \sum_{k=1}^{K} \alpha_{t,k}\mathbf{W}_r^{(k)}, \mathbf{W}_h^{(t)} = \sum_{k=1}^{K} \alpha_{t,k}\mathbf{W}_h^{(k)}. \tag{17}$$

This design allows the PDD-GRU to adapt its transition dynamics to the current visit pattern by retrieving the most relevant evolution behavior from the expert pool.

Finally, we integrate the structural variation and dynamic weights into the recurrent update. We introduce an evolution gate $\gamma_t \in (0,1)^d$ to control how much the hidden state should be updated based on the severity of dependency changes:

$$\gamma_t = \sigma(\mathbf{W}_{prog}\mathbf{v}_{prog,t} + \mathbf{b}_{prog}), \tag{18}$$

where $\mathbf{W}_{prog} \in \mathbb{R}^{d\times 3}$ and $\mathbf{b}_{prog} \in \mathbb{R}^d$ are learnable parameters. The update equations are:

$$\begin{aligned}
\mathbf{z}_t &= \sigma(\mathbf{W}_z^{(t)}\mathbf{g}_t + \mathbf{U}_z\mathbf{h}_{t-1}),\\
\mathbf{r}_t &= \sigma(\mathbf{W}_r^{(t)}\mathbf{g}_t + \mathbf{U}_r\mathbf{h}_{t-1}),\\
\tilde{\mathbf{h}}_t &= \tanh(\mathbf{W}_h^{(t)}\mathbf{g}_t + \mathbf{U}_h(\mathbf{r}_t \odot \mathbf{h}_{t-1})),\\
\tilde{\mathbf{h}}_c &= (1-\mathbf{z}_t)\odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t,\\
\mathbf{h}_t &= (1-\gamma_t)\odot \mathbf{h}_{t-1} + \gamma_t \odot \tilde{\mathbf{h}}_c,
\end{aligned} \tag{19}$$

where $\mathbf{U}_*$ represents the static recurrent weights. This design enables the PDD-GRU to preserve historical memory under stable disease states while rapidly adapting to visits exhibiting strong structural shifts.

4.3 Recommendation

To generate the final medication recommendation for the last visit t, we fuse sequence-level historical context and visit-level structural information. Specifically, we concatenate the PDD-GRU hidden state $\mathbf{h}_t$ with the hypergraph readout vector $\mathbf{g}_t$. This ensures that the prediction is conditioned on both long-term history and immediate structural features. The fused representation is passed through a linear classifier to compute the medication probabilities:

$$\mathbf{e}_{fuse} = [\mathbf{h}_t \parallel \mathbf{g}_t], \tag{20}$$

$$\hat{\mathbf{y}}_t = \sigma(\mathbf{W}_{out}\cdot \mathbf{e}_{fuse} + \mathbf{b}_{out}), \tag{21}$$

where $\mathbf{W}_{out} \in \mathbb{R}^{|\mathcal{M}|\times 2d}$ and $\mathbf{b}_{out} \in \mathbb{R}^{|\mathcal{M}|}$ are learnable parameters, and $\sigma(\cdot)$ is the Sigmoid activation function.

4.4 Training and Optimization

The training of SpecTD-MR is conducted in two stages: hypergraph pretraining in Section 4.1 and downstream post-training. In the post-training stage, we initialize the sequence encoder using the pretrained hypergraph representations and visit prototypes. Since clustering is a full-graph operation, its objective is applied only during pretraining. To prevent the learned disease spectra from drifting while still accommodating the newly introduced modules, we apply a smaller learning rate to the Time-evolving Dependency Modeling module than to the Disease Spectrum-aware Hypergraph Learning module. The entire framework is then optimized end-to-end for the multi-label medication recommendation task.

Just Accepted

Table 2. Performance Comparison. The best results are highlighted in bold, and the second-best results are underlined. All improvements are statistically significant with a p-value < 0.05.

Methods	MIMIC-III			MIMIC-IV		
	Jaccard	F1-score	PRAUC	Jaccard	F1-score	PRAUC
LR	0.4788 ± 0.0041	0.6371 ± 0.0040	0.7401 ± 0.0044	0.4348 ± 0.0020	0.5824 ± 0.0022	0.7061 ± 0.0018
ECC	0.4730 ± 0.0029	0.6326 ± 0.0029	0.7492 ± 0.0035	0.4109 ± 0.0030	0.5537 ± 0.0031	0.7037 ± 0.0018
RETAIN	0.4793 ± 0.0028	0.6411 ± 0.0026	0.7606 ± 0.0041	0.4161 ± 0.0017	0.5701 ± 0.0019	0.6721 ± 0.0021
LEAP	0.4179 ± 0.0019	0.5728 ± 0.0019	0.6757 ± 0.0021	0.4063 ± 0.0016	0.5589 ± 0.0015	0.5216 ± 0.0017
GAMENet	0.4965 ± 0.0037	0.6543 ± 0.0033	0.7531 ± 0.0048	0.4557 ± 0.0019	0.6089 ± 0.0017	0.7104 ± 0.0013
SafeDrug	0.4990 ± 0.0038	0.6577 ± 0.0034	0.7482 ± 0.0041	0.4549 ± 0.0024	0.6075 ± 0.0023	0.6929 ± 0.0017
MICRON	0.5209 ± 0.0023	0.6746 ± 0.0020	0.7554 ± 0.0021	0.4515 ± 0.0007	0.6046 ± 0.0006	0.6676 ± 0.0008
COGNet	0.5189 ± 0.0042	0.6712 ± 0.0040	0.6944 ± 0.0038	0.4801 ± 0.0027	0.6282 ± 0.0024	0.6379 ± 0.0023
MoleRec	0.5153 ± 0.0037	0.6720 ± 0.0033	0.7598 ± 0.0042	0.4466 ± 0.0024	0.6008 ± 0.0025	0.6892 ± 0.0025
VITA	0.5211 ± 0.0012	0.6751 ± 0.0012	0.7609 ± 0.0013	0.4612 ± 0.0007	0.6089 ± 0.0007	0.6677 ± 0.0006
ARMR	0.5307 ± 0.0011	0.6858 ± 0.0023	0.7801 ± 0.0025	0.4910 ± 0.0013	0.6389 ± 0.0019	0.7315 ± 0.0020
MR-DTR	0.5416 ± 0.0041	0.6935 ± 0.0038	0.7763 ± 0.0035	0.4921 ± 0.0019	0.6406 ± 0.0018	0.7265 ± 0.0022
SpecTD-MR	**0.5514 ± 0.0018**	**0.7033 ± 0.0024**	**0.7877 ± 0.0023**	**0.5037 ± 0.0021**	**0.6531 ± 0.0025**	**0.7379 ± 0.0031**

Let $\mathbf{y}_t \in \{0,1\}^{|\mathcal{M}|}$ denote the ground-truth medication multi-hot vector for the last visit. We combine multi-label binary cross-entropy loss and multi-label margin loss [10] to jointly minimize the discrepancy between predictions and ground truth:

$$\mathcal{L}_{\text{med}} = -\sum_{i=1}^{|\mathcal{M}|}\Big[\mathbf{y}_{t,i}\log\hat{\mathbf{y}}_{t,i} + (1-\mathbf{y}_{t,i})\log(1-\hat{\mathbf{y}}_{t,i})\Big] + \beta \sum_{i:\,\mathbf{y}_{t,i}=1}\sum_{j:\,\mathbf{y}_{t,j}=0}\frac{\max\big(0,\,1-(\hat{\mathbf{y}}_{t,i}-\hat{\mathbf{y}}_{t,j})\big)}{|\mathcal{M}|}, \tag{22}$$

where β controls the contribution of the margin-based ranking constraint. During inference, a medication i is recommended if its predicted probability $\hat{\mathbf{y}}_{t,i}$ exceeds a threshold of 0.5.

5 Experiments

5.1 Experimental Setup

5.1.1 Datasets and Data Processing. Our experiments were conducted on two of the most widely used datasets, MIMIC-III [17] and MIMIC-IV [16]. During the data preprocessing phase, we adopt a rigorous pipeline to transform raw clinical records into structured longitudinal sequences. To ensure the temporal integrity of the clinical trajectories, we sort all admissions for each patient chronologically based on their admission timestamps. This step guarantees that the data reflects a valid, forward-moving disease progression, which is fundamental for modeling time-evolving dependencies. Clinical entities were encoded using standardized taxonomies: diagnoses and procedures were mapped via the International Classification of Diseases, Ninth Revision (ICD-9), while medications were categorized according to the third level of the Anatomical Therapeutic Chemical (ATC) classification system. Utilizing the ATC-3 level allows for the aggregation of individual drugs into specific pharmacological or therapeutic subgroups, enhancing the model's ability to capture generalized patterns. The study cohort was restricted to patients with a minimum of two clinical encounters to facilitate longitudinal analysis. Furthermore, to model adverse drug-drug interactions (DDIs), we adopted the methodology proposed by SafeDrug [43]

Just Accepted

Table 3. The Data Statistics of MIMIC-III/IV Datasets

Items	MIMIC-III	MIMIC-IV
# of visits / # of patients	14,124 / 5,442	149,561 / 44,937
space size of diagnoses / procedures	1,956 / 1,407	1,992 / 1,999
space size of drugs	120	125
avg. / max # of visits	2.60 / 29	3.33 / 70
avg. / max # of diagnoses per visit	10.36 / 128	7.97 / 267
avg. / max # of procedures per visit	3.85 / 50	2.18 / 70
avg. / max # of drugs per visit	11.21 / 64	6.21 / 72
# of DDI types	40	40
total # of DDI pairs	350	981

and extracted DDI information from the DrugBank database [35]. In alignment with established benchmarks [5, 30, 43, 44], the DDI graph was constructed using the standard configuration of the top 40 most frequent interaction types at the ATC-3 level.

5.1.2 Implementation Details. Following the settings commonly adopted in prior work [22, 43, 44], we split each dataset into training, validation, and testing sets with a ratio of 4:1:1. All embedding dimensions, including those for concepts, hyperedge, the hyper convolution layer, the transformer block uniformly set to 128. The learning rate for Time-evolving Dependency Modeling is set to 5e-4, while the learning rate for the Disease Spectrum-aware Hypergraph Learning module is set to 1e-3 during the pre-training phase and 1e-5 otherwise. The loss-function hyperparameters φ and η were set to 1.0 and 0.1, respectively. Set the number of clusters K to 5. In the Disease Spectrum-aware Hypergraph Learning phase, hypergraph construction and pretraining are performed solely using the training set. No information from the validation or test set is used during pretraining, ensuring the reproducibility and fairness of the experimental setup. In the subsequent Time-evolving Dependency Modeling phase, patients from the validation and test sets are dynamically incorporated into the hypergraph (excluding medications from their final visits to prevent label leakage), utilizing pre-trained hyperedge and node embeddings for rapid initialization. We implemented our model in PyTorch 2.1 and optimized all parameters using the Adam optimizer [20]. The source code has been released for reproducibility. All experiments were conducted on a cluster equipped with four NVIDIA GeForce RTX 4090 GPUs. Baseline models were reproduced using the official implementations and recommended configurations provided by previous work [22, 39, 43, 44].

5.1.3 Baselines. We evaluate the effectiveness of our method by comparing it against 12 representative and state-of-the-art baseline methods:

- **Logistic Regression (LR)** is a classical instance-based standard classifier with L2 regularization.
- **Ensembles of Classifier Chains (ECC)** [28] utilizes multiple SVM classifiers for multi-label prediction.
- **RETAIN** [7] captures longitudinal patient information using a two-level neural attention mechanism.
- **LEAP** [50] is a generative recommendation model based on LSTM [12].
- **GAMENet** [30] incorporates a drug–drug interaction knowledge graph via a memory-augmented neural network.
- **SafeDrug** [43] employs a dual molecular graph encoder to jointly learn global and local structural representations of drugs.
- **MICRON** [42] iteratively updates hidden drug representations and recommended drug sets through a reconstructive design.
- **COGNet** [39] computes drug replication probabilities as auxiliary guidance for medication prediction.

Just Accepted

- **MoleRec** [44] models interactions among molecular substructures and their dependencies with patient health conditions.
- **VITA** [19] selectively retrieves past visits most relevant to the current visit and accurately measures visit-level relevance.
- **ARMR** [36] employs adaptive response modules to determine how to utilize existing and new drugs.
- **MR-DTR** [22] effectively learns drug characterization using guidance information from similar patients.

5.1.4 Evaluation Metrics. Following prior studies in medication recommendation [22, 30, 44], we employ four commonly used evaluation metrics: Jaccard Similarity Score (Jaccard), Average F1-score (F1) and Precision–Recall AUC (PRAUC). Formally, given the test patient set $\mathcal{U}$, we predict the last visit result for each patient, we employ the following metrics:

- **Jaccard**:

$$\text{Jaccard} = \frac{1}{|\mathcal{U}|} \sum_{i=1}^{|\mathcal{U}|} \frac{|\mathbf{y}_i \cap \hat{\mathbf{y}}_i|}{|\mathbf{y}_i \cup \hat{\mathbf{y}}_i|}. \tag{23}$$

- **F1**:

$$\begin{aligned} \text{Precision}_i = \frac{|\mathbf{y}_i \cap \hat{\mathbf{y}}_i|}{|\hat{\mathbf{y}}_i|}, \quad \text{Recall}_i = \frac{|\mathbf{y}_i \cap \hat{\mathbf{y}}_i|}{|\mathbf{y}_i|}, \\ \text{F1} = \frac{1}{|\mathcal{U}|} \sum_{i=1}^{|\mathcal{U}|} \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}. \end{aligned} \tag{24}$$

- **PRAUC**:

$$\text{PRAUC} = \frac{1}{|\mathcal{U}|} \sum_{i=1}^{|\mathcal{U}|} \sum_{k=1}^{|\mathcal{M}|} \text{Precision}(k)_i \cdot \Delta\,\text{Recall}(k)_i, \tag{25}$$

where k represents the rank in the predicted probability distribution, $\text{Precision}(k)_i$ is the precision at cut-off k, and $\Delta\,\text{Recall}(k)_i = \text{Recall}(k)_i - \text{Recall}(k-1)_i$ denotes the change in recall.

5.2 Result Analysis

Table 2 presents the experimental results of SpecTD-MR compared with all baseline models. We evaluate each method using 10 rounds of bootstrap sampling with sampling rate = 0.8 and report the mean and standard deviation of all metrics. We conducted a paired t-test to assess statistical significance. For each dataset and each evaluation metric, SpecTD-MR was compared with each baseline method, and all improvements achieved $p < 0.05$. Overall, longitudinal models such as GAMENet, SafeDrug, VITA, ARMR, and MR-DTR substantially outperform traditional approaches (LR and ECC) and instance-based methods (LEAP). Neural network–based models designed specifically for medication recommendation are capable of capturing more realistic clinical reasoning processes and richer feature interactions. Moreover, by leveraging the longitudinal trajectories in EHRs to characterize a patient's evolving health conditions, these models reveal the limitations of instance-based methods, which exhibit a clear trend toward obsolescence. Early deep learning models such as RETAIN, GAMENet, and SafeDrug rely on simple RNN-based architectures to model complex longitudinal histories, which restricts their capacity to capture nuanced temporal patterns. Transformer-based methods, including COGNet and state-space-driven ARMR, achieve notable improvements but remain constrained by their coarse-grained representations of patient health status. MR-DTR incorporates dynamic treatment regimens, leading to further performance gains. However, its reliance on only coarse-grained similar patient information results in suboptimal performance compared to our approach.

Just Accepted

Table 4. Comparison of SpecTD-MR with its variants

Variants	MIMIC-III			MIMIC-IV		
	Jaccard	F1-score	PRAUC	Jaccard	F1-score	PRAUC
SpecTD-MR *w/o DSHL*	0.5302 ± 0.0035	0.6810 ± 0.0038	0.7655 ± 0.0036	0.4835 ± 0.0029	0.6302 ± 0.0031	0.7170 ± 0.0034
SpecTD-MR *w/o CCL*	0.5370 ± 0.0032	0.6890 ± 0.0034	0.7730 ± 0.0030	0.4921 ± 0.0027	0.6395 ± 0.0028	0.7286 ± 0.0031
SpecTD-MR *w/o Mask*	0.5420 ± 0.0028	0.6945 ± 0.0030	0.7805 ± 0.0029	0.4978 ± 0.0025	0.6450 ± 0.0026	0.7345 ± 0.0028
SpecTD-MR *w/o HHDI*	0.5285 ± 0.0031	0.6715 ± 0.0033	0.7558 ± 0.0032	0.4788 ± 0.0024	0.6322 ± 0.0027	0.7201 ± 0.0029
SpecTD-MR *w/o* A^{prior}	0.5452 ± 0.0024	0.6981 ± 0.0027	0.7830 ± 0.0026	0.4961 ± 0.0025	0.6449 ± 0.0028	0.7315 ± 0.0030
SpecTD-MR *w/o PDD*	0.5450 ± 0.0026	0.6972 ± 0.0029	0.7832 ± 0.0027	0.4990 ± 0.0023	0.6470 ± 0.0025	0.7352 ± 0.0028
SpecTD-MR *w/* Trans-ICD	0.5324 ± 0.0024	0.6824 ± 0.0029	0.7693 ± 0.0032	0.4866 ± 0.0059	0.6371 ± 0.0031	0.7167 ± 0.0027
SpecTD-MR *w/* Temp-Trans	0.5417 ± 0.0026	0.6904 ± 0.0047	0.7791 ± 0.0034	0.4903 ± 0.0025	0.6411 ± 0.0039	0.7309 ± 0.0024
SpecTD-MR *w/* GRU-Attn	0.5443 ± 0.0021	0.6914 ± 0.0025	0.7806 ± 0.0029	0.4971 ± 0.0059	0.6436 ± 0.0037	0.7314 ± 0.0035
SpecTD-MR	**0.5514 ± 0.0018**	**0.7033 ± 0.0024**	**0.7877 ± 0.0023**	**0.5037 ± 0.0021**	**0.6531 ± 0.0025**	**0.7379 ± 0.0031**

5.3 Ablation Study

We assess the contribution of each component in SpecTD-MR through ablation studies. Several model variants are constructed by removing or modifying specific modules: SpecTD-MR *w/o DSHL* excludes the Disease Spectrum-aware Hypergraph Learning module and uses random initialization, SpecTD-MR *w/o CCL* removes the clustering and contrastive losses, SpecTD-MR *w/o Mask* eliminates the masked prediction warm-up, SpecTD-MR *w/o HHDI* replaces the History-aware Hidden Dependency Inference mechanism with direct summation of concept embeddings, SpecTD-MR *w/o PDD* substitutes the PDD-GRU with a vanilla GRU, and SpecTD-MR *w/o* A^{prior} removes the prior co-occurrence matrix A^{prior}. In addition to component ablations, we introduce three complexity-controlled alternatives to further verify the necessity and effectiveness of the proposed architecture. SpecTD-MR *w/* Trans-ICD replaces DSHL with an ICD/ATC-initialized Transformer encoder. SpecTD-MR *w/* Temp-Trans replaces PDD-GRU and MoE-based transition with a positional temporal Transformer, while retaining the pretrained concept representations and HHDI. SpecTD-MR *w/* GRU-Attn replaces the proposed temporal association modeling with a parameter-matched GRU plus visit-level attention.

Table 4 summarizes the performance of these variants. The performance drop in SpecTD-MR *w/o DSHL* confirms the importance of spectrum-aware representations and external knowledge. The degradation in SpecTD-MR *w/o CCL* highlights the necessity of task-guided clustering and alignment. The weaker results of SpecTD-MR *w/o Mask* demonstrate that warm-up pretraining is crucial for stabilizing spectrum learning. The decline observed in SpecTD-MR *w/o HHDI* shows the need for explicit modeling of inter-visit dependencies. SpecTD-MR *w/o* A^{prior} still substantially outperforms SpecTD-MR *w/o HHDI*, indicating that the history-aware fusion mechanism remains the major contributor, while A^{prior} serves as a complementary regularizer that stabilizes dependency estimation with population-level clinical co-occurrence patterns. Finally, SpecTD-MR *w/o PDD* reveals that generic sequential encoders are less effective in capturing the evolution of disease states, resulting in inferior recommendations. These three variants further verify that the performance improvements of SpecTD-MR cannot be simply attributed to increased model capacity or the use of standard neural network architectures. SpecTD-MR *w/* Trans-ICD performs worse than the full model, indicating that ICD/ATC-initialized Transformer encoding cannot fully replace disease spectrum-aware hypergraph learning. SpecTD-MR *w/* Temp-Trans also lags behind SpecTD-MR, suggesting that a standard temporal Transformer is insufficient to model association-variation-aware disease evolution. In addition, SpecTD-MR *w/* GRU-Attn remains inferior even with comparable parameter size,

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Table 5. Frequency-stratified analysis of A^{prior} on MIMIC-III.

Group	Model	Jaccard	F1-score	PRAUC
Low	w/o A^{prior}	0.5549± 0.0026	0.7062± 0.0037	0.7892± 0.0031
Low	w/ A^{prior}	0.5702± 0.0032	0.7183± 0.0040	0.8032± 0.0039
Medium	w/o A^{prior}	0.5445± 0.0024	0.6988± 0.0036	0.7867± 0.0043
Medium	w/ A^{prior}	0.5498± 0.0038	0.7021± 0.0028	0.7870± 0.0023
High	w/o A^{prior}	0.5362± 0.0020	0.6893± 0.0027	0.7731± 0.0018
High	w/ A^{prior}	0.5341± 0.0035	0.6894± 0.0044	0.7728± 0.0025

demonstrating that the gains cannot be explained by model capacity alone. These results support the usefulness of the proposed disease spectrum-aware representation learning and time-evolving association modeling.

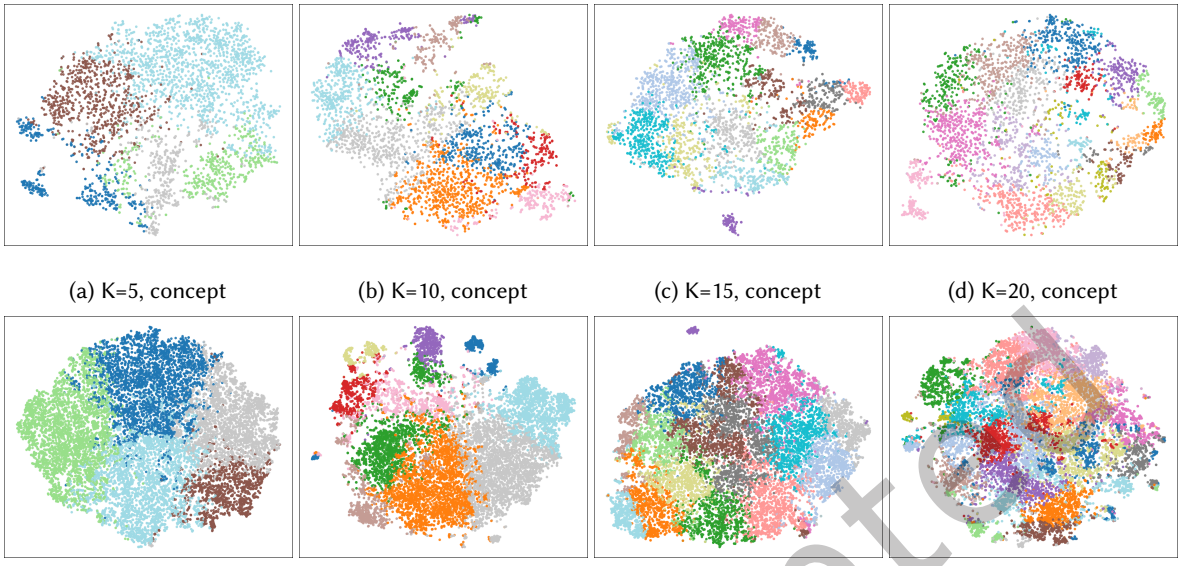
To further examine the behavior of the population-level prior on rare diagnoses, we conduct a frequency-stratified analysis on the MIMIC-III test set. Different from single-label diagnosis prediction, medication recommendation is performed at the visit level, where each visit contains a set of diagnosis codes. Therefore, an individual rare diagnosis cannot be directly isolated as an independent test instance. Inspired by RareMed [53], we stratify test samples according to the average occurrence frequency of diagnosis codes in the last visit. Specifically, we divide the test samples into three equal-sized groups, corresponding to low-, medium-, and high-frequency regimes. The diagnosis frequency ranges of these groups are 2–15, 15–32, and 33–376, with mean frequencies of 9.99, 22.44, and 80.90, respectively. It should be noted that the high-frequency group only indicates that the diagnosis codes are more common in the dataset, and does not necessarily imply that the medication combination is easier to predict. As shown in Table 5, incorporating A^{prior} does not degrade performance on the low-frequency group; instead, it slightly improves Jaccard, F1, and PRAUC, suggesting that the population-level prior can provide useful auxiliary signals for rare diagnoses. For the medium-frequency group, the accuracy metrics remain nearly unchanged. For the high-frequency group, however, A^{prior} leads to a decrease in accuracy, indicating that population-level co-occurrence patterns may over-regularize common diagnoses and bias the model toward more frequent treatment patterns. These results indicate that A^{prior} is generally helpful or neutral for low- and medium-frequency diagnoses, while its effect may vary across diagnosis-frequency regimes. This observation also motivates future work on frequency-adaptive prior weighting.

5.4 Case Study

We present representative examples from the MIMIC-IV test set to illustrate the interpretability and effectiveness of SpecTD-MR.

5.4.1 Clustering Analysis. Figure 3 visualizes the latent representations across varying granularities ($K \in \{5, 10, 15, 20\}$). We initially observe that the topological structure of the visit space mirrors that of the concept space, confirming that SpecTD-MR successfully bridges the semantic gap via contrastive alignment. Beyond this structural consistency, the clusters exhibit a natural hierarchical evolution: as K increases, broad therapeutic disease spectra are further separated into finer-grained treatment-related patterns, suggesting a hierarchical organization in the learned representation space. For instance, at $K = 5$, the model separates a *Chronic Metabolic Spectrum*—linking *diabetes*-related diagnoses with maintenance medications (e.g., C09A)—from an *Acute Spectrum* containing complications such as *myocardial infarction*. This learned semantic proximity provides an alternative explanation for why SpecTD-MR subsequently recommends renal protective agents (C09A) for progressive diabetic conditions.

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(a) K=5, concept (b) K=10, concept (c) K=15, concept (d) K=20, concept

(e) K=5, visit (f) K=10, visit (g) K=15, visit (h) K=20, visit

Fig. 3. t-SNE visualization of learned representations for Concepts and Visits with varying cluster numbers K.

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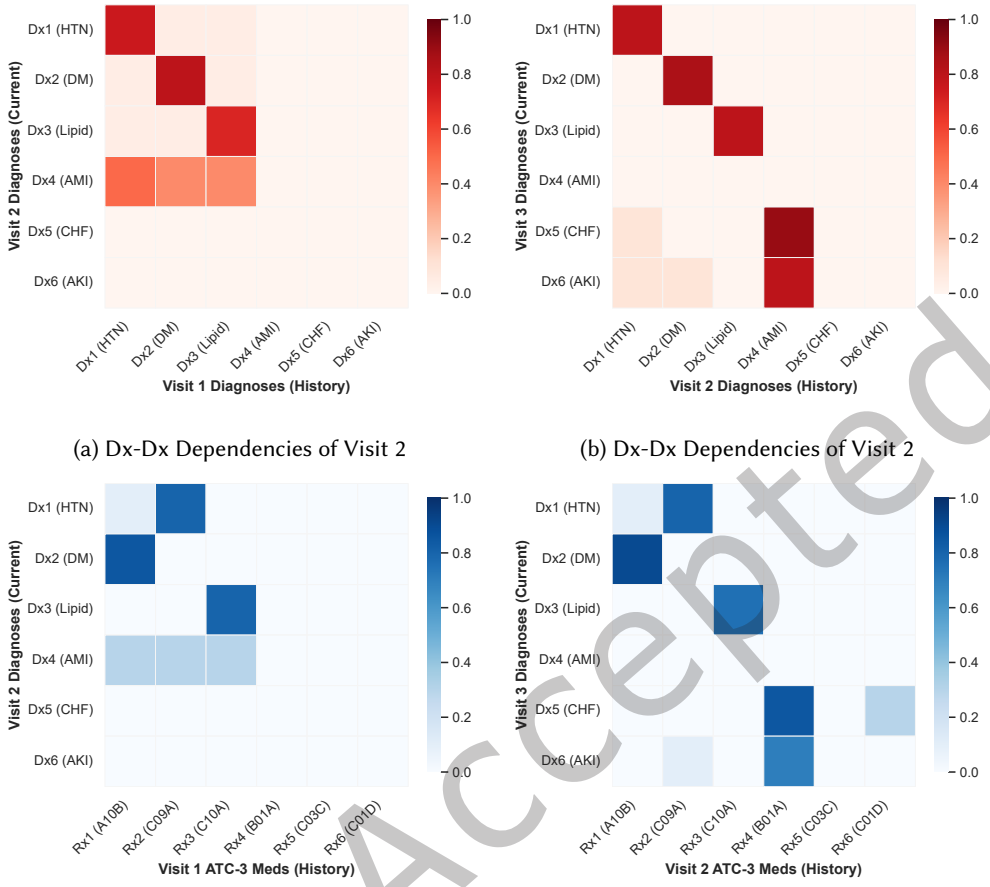
As an auxiliary reference, we further compare the learned clusters with ICD-9 chapter groupings using Purity, NMI, and ARI when $K = 20$. The results are Purity = 0.335, NMI = 0.211, and ARI = 0.105. These values indicate partial alignment with ICD chapter-level anatomical groupings, while also showing clear divergence from strict taxonomic boundaries. This is consistent with the intended role of our clustering module, which is optimized for therapeutic similarity rather than pathophysiological taxonomy. Therefore, ICD-based metrics are reported as a supplementary reference rather than the primary evaluation criterion for the learned therapeutic spectra. The effectiveness of these therapeutic spectra is quantitatively validated by their substantial contribution to downstream recommendation performance and the clinically coherent drug adaptations shown in the case studies.

5.4.2 Time-evolved Dependencies. Figure 4 visualizes dependency matrices across three visits of a patient. In Visit 2, Dx4 (*Acute Myocardial Infarction*) attends strongly to Dx1–Dx3 (*hypertension, diabetes, and lipid disorders*), indicating that SpecTD-MR identifies these chronic diseases as upstream risk factors. In Visit 3, Dx5 (*Heart Failure*) focuses primarily on its counterpart in Visit 2, capturing the cascade of cardiac deterioration. Notably, Chronic diagnoses consistently assign high attention to themselves, reflecting their stable self-dependent progression patterns.

Medication-level dependencies also emerge: Dx2 (*diabetes*) attends heavily to Rx1 (A10B), indicating correct modeling of long-term medication stability. Dx4 retains attention to chronic-disease medications, aligning with their frequent co-occurrence. Finally, Dx5's attention to Rx4 (B01A) reflects recognition of medications as signals of prior processes rather than causal triggers.

It is important to note that these visualizations are intended as qualitative sanity checks and supportive case studies rather than definitive evidence of causal interpretability. They demonstrate how the model attends to historical clinical contexts without claiming rigorous mechanistic reasoning.

5.4.3 Result Comparison. As shown in Table 6, the upper section presents the longitudinal evolution of diagnoses, while the lower section compares the medication recommendations for the third visit. In this scenario, the diabetes



(a) Dx-Dx Dependencies of Visit 2

(b) Dx-Dx Dependencies of Visit 2

(c) Dx-Rx Dependencies of Visit 2

(d) Dx-Rx Dependencies of Visit 3

Fig. 4. Four dependency submatrices of cross-visit patterns.

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diagnosis shifts from the previously observed 250.00 (*Type 2 Diabetes, without complications*) and 250.60 *(Type 2 Diabetes, with neurological manifestations*) to the less common 250.40 (*Diabetes with renal manifestations*). MR-DTR accurately recommends the A10B, owing to its ability to consider dynamic treatment trajectories from clinically similar patients. COGNet succeeds because it replicates medications that have appeared frequently in prior visits, whereas MoleRec, which employs independent code embeddings, fails to capture this relationship. In contrast, our SpecTD-MR model effectively learns the disease hierarchy, enabling it to transfer medication patterns within the same diagnostic family and correctly recommend the A10B drug.

A more challenging shift occurs when the patient develops 585.3 (*Chronic Kidney Disease*). MR-DTR and MoleRec recommend C03C, a *general diuretic*, focusing only on the current renal diagnosis. In contrast, our SpecTD-MR model recognizes CKD as a downstream effect of diabetes and hypertension and recommends C09A, which aligns with clinical guidelines for renal protection in diabetic patients. These findings demonstrate SpecTD-MR's ability to jointly model disease spectra and time-evolving dependencies for clinically coherent recommendations.

Table 6. An example illustrating the necessity of disease spectrum-aware and time-evolving dependency learning.

Visit 1 & 2 (History Context)					
• V1: 250.00 (T2 DM), 401.9 (HTN) → Rx: A10B, C09A					
• V2: 250.60 (DM Neuro), 272.4 (Lipid) → Rx: A10B, C09A, C10A					
Visit 3 (Prediction Target)					
Dx Input: 250.40 (DM Renal), **585.3** (CKD III)					
**Note the code shift (250.00→250.40) & progression (CKD).*					
Comparison of Recommendation Results					
Drug	**GT**	**MR-DTR**	**MoleRec**	**COGNet**	**Ours**
A10B	✓	✓	-	✓	✓
→ Handling hierarchy shift (250.00 → 250.40)					
C09A	✓	-	-	-	✓
C03C	✗	✗	✗	-	-
→ Modeling dynamic evolution (DM → CKD)					
C10A	✓	✓	✓	✓	✓

5.5 Parameter Analysis

In this subsection, we systematically investigate the sensitivity of SpecTD-MR to eight key hyperparameters in order to assess its robustness. These hyperparameters include the cluster number K, the projection and hidden dimensions, the number of attention heads, the learning rates for the two training stages, and the weights for the clustering and alignment losses. We selected representative configurations for these hyperparameters, conducted experiments on the MIMIC-III and MIMIC-IV datasets, and used the Jaccard metric to quantify prediction accuracy, as shown in Figure 5.

To begin with, the cluster number K determines the granularity of the disease spectrum; performance peaks at $K = 5$, validating the necessity of capturing distinct disease subtypes rather than a monolithic representation. While higher K values can reveal finer-grained treatment-oriented spectrum patterns, this excessive granularity introduces semantic fragmentation for the downstream recommendation task. Methodologically, forcing the model to separate clinically similar visits into highly specific disjoint clusters reduces generalization and exacerbates data sparsity. Clinically, since our prediction target operates at the ATC-3 level (broad pharmacological subgroups), broader disease spectra ($K = 5$) provide a better inductive bias by successfully aligning macroscopic latent therapeutic disease spectra with macroscopic treatment strategies, thereby yielding the highest recommendation accuracy. In parallel, increasing the model capacity—specifically the projection and hidden dimensions to 128—yields consistent gains by enabling the encoder to fully integrate rich multi-source knowledge; however, further expanding these dimensions introduces sparsity and noise, leading to overfitting where the model memorizes training samples rather than learning robust features. Similarly, using 4 attention heads achieves optimal performance by allowing the model to attend to diverse semantic subspaces simultaneously, whereas more heads disperse attention weights too thinly.

Furthermore, the optimization strategy proves critical for stability. The pre-training stage requires a larger learning rate to effectively drive the initial organization of the hypergraph structure and help the model escape local minima. In contrast, the downstream recommendation stage demands a reduced learning rate to fine-tune the sequence encoder, ensuring that the structural knowledge acquired during pre-training is adapted without being catastrophically overwritten. Finally, the loss weights for auxiliary tasks exhibit a clear sweet spot; weights

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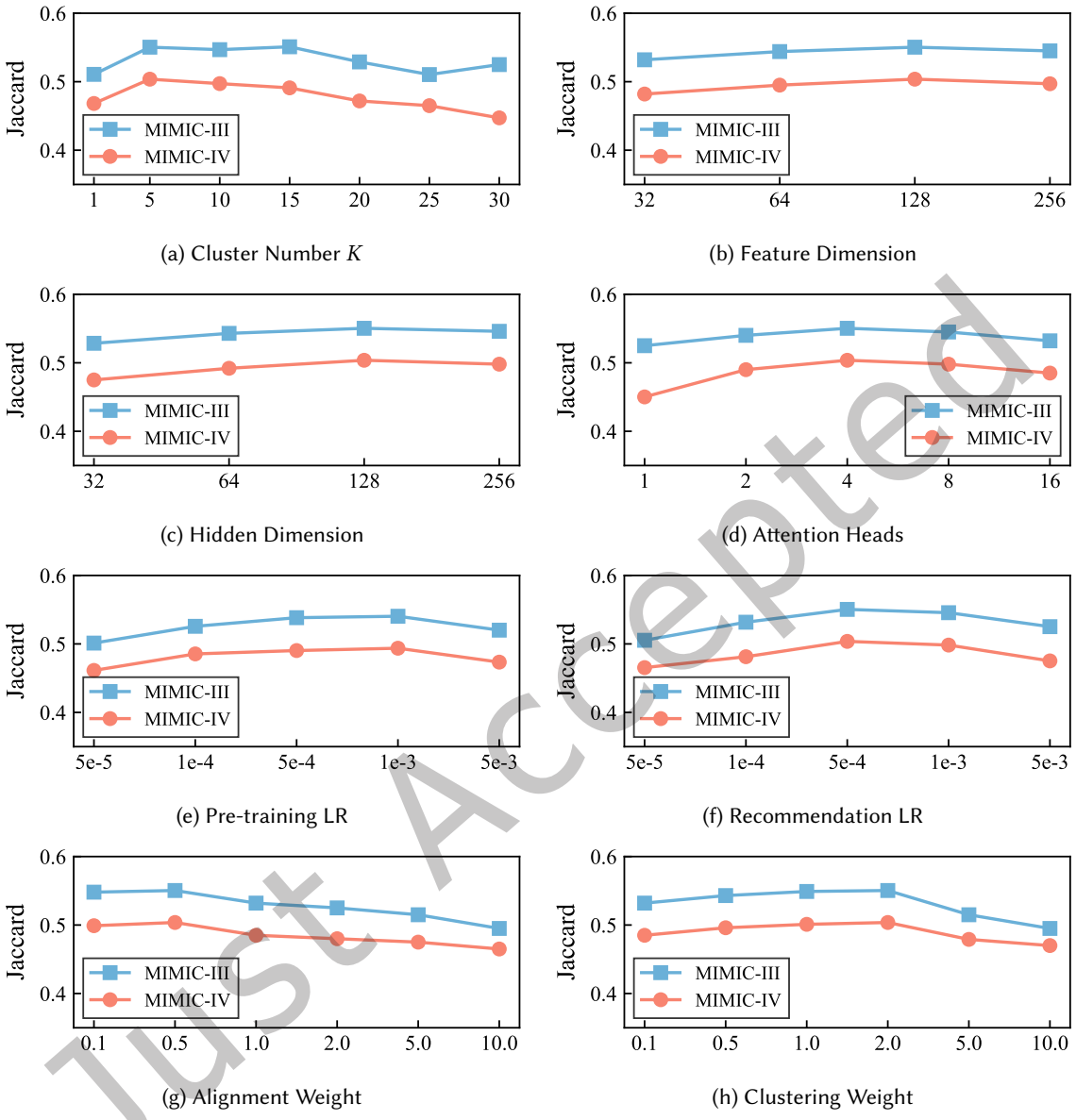


Fig. 5. Impact of key hyperparameters on SpecTD-MR.

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that are too low provide insufficient structural guidance to bridge the semantic gap, while excessively high weights cause auxiliary objectives to dominate gradient updates, distracting the model from the primary medication recommendation task.

Table 7. Model Complexity and Training/Inference Efficiency on MIMIC-III.

Method	# Parameters	Training Time (s/epoch)	Inference Time (ms/sample)
LEAP	479,944	558.75	10.12
GAMENet	886,662	132.40	3.58
MICRON	712,965	111.88	2.25
COGNet	3,192,910	290.34	17.19
VITA	2,897,715	331.45	18.06
SpecTD-MR	1,774,844	415.72	5.13

5.6 Efficiency Analysis

We analyze the computational efficiency of SpecTD-MR in terms of model size, training time, and online inference time, and compare it with representative baselines on the MIMIC-III dataset. Table 7 reports the number of trainable parameters, the average training time per epoch, and the average per-sample inference time on the test set under the same experimental environment. The inference time is measured with batch size 1, where each patient is processed independently to simulate online clinical recommendation. Overall, SpecTD-MR exhibits a moderate model size, containing approximately 1.77M parameters. While it is larger than lightweight sequential models such as LEAP and MICRON, it remains substantially more compact than graph-heavy or highly parameterized architectures such as COGNet and VITA. This reflects a balanced design choice: SpecTD-MR introduces additional structural components (e.g., hypergraph encoding and prototype-driven dependency modeling) without incurring excessive parameter growth. In terms of training efficiency, SpecTD-MR requires more training time per epoch than purely sequential or DDI-oriented models, which is expected due to the additional cost of hypergraph message passing and history-aware dependency inference. Nevertheless, its training time remains comparable to other structure-aware baselines and significantly lower than that of large graph-based models.

The relatively higher training cost mainly comes from the two-stage learning procedure, where the hypergraph encoder is pretrained to learn disease spectrum-aware representations before downstream recommendation training. Since this additional cost is incurred offline, it does not directly affect the online response time during deployment. For online inference, SpecTD-MR achieves an average time of 5.13 ms per sample. While its inference time exceeds that of lightweight models like GAMENet and MICRON, it remains at the millisecond level and is significantly lower than complex baselines such as COGNet and VITA. This indicates that the additional modeling capacity of SpecTD-MR does not introduce prohibitive online latency. During inference, the model does not repeat the hypergraph pretraining or clustering process; instead, it only uses the learned disease spectrum-aware representations to perform history-aware dependency inference, prototype routing, and recurrent state updating for the target patient. Importantly, the added computational cost directly corresponds to the modeling of disease spectrum structures and time-evolving dependencies, which are shown to substantially improve recommendation accuracy and interpretability in previous sections. Therefore, the proposed framework achieves a favorable trade-off between modeling capacity and computational efficiency, making it practical for real-world clinical decision support scenarios where offline training is acceptable and online recommendation needs to be completed within a short response time.

5.7 DDI Discussion

Although DDI is a critical safety concern in medication recommendation, our primary goal in this work is to investigate how disease spectrum awareness and time-evolving dependencies improve the clinical coherence of

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Table 8. DDI Rate Comparison (Lower is Better) with Improvement over the Real DDI Rate.

Method	MIMIC-III	Δ vs. Real	MIMIC-IV	Δ vs. Real
Ground-truth (observed in test set)				
Real DDI Rate	0.0790	–	0.0743	–
DDI-aware baselines (explicitly model / regularize DDI)				
SafeDrug	**0.0642 ± 0.0003**	+0.0148	**0.0552 ± 0.0002**	+0.0191
MICRON	<u>0.0683 ± 0.0003</u>	+0.0107	<u>0.0572 ± 0.0002</u>	+0.0171
MoleRec	0.0725 ± 0.0005	+0.0065	0.0605 ± 0.0004	+0.0138
MR-DTR	0.0707 ± 0.0006	+0.0083	0.0651 ± 0.0004	+0.0092
General medication recommendation models				
LR	0.0777 ± 0.0011	+0.0013	0.0658 ± 0.0006	+0.0085
ECC	0.0745 ± 0.0015	+0.0045	0.0656 ± 0.0005	+0.0087
RETAIN	0.0849 ± 0.0015	−0.0059	0.0807 ± 0.0012	−0.0064
LEAP	0.0760 ± 0.0013	+0.0030	0.0620 ± 0.0007	+0.0123
GAMENet	0.0818 ± 0.0006	−0.0028	0.0775 ± 0.0003	−0.0032
COGNet	0.0766 ± 0.0012	+0.0024	0.0755 ± 0.0003	−0.0012
VITA	0.0750 ± 0.0003	+0.0040	0.0761 ± 0.0003	−0.0018
ARMR	0.0758 ± 0.0019	+0.0032	0.0688 ± 0.0015	+0.0055
SpecTD-MR	0.0768 ± 0.0010	+0.0022	0.0693 ± 0.0008	+0.0050

Note: Δ vs. Real is computed as (Real DDI Rate − Predicted DDI Rate). Positive values indicate reductions in DDI rate compared with the observed test-set DDI rate. Bold and underlined numbers indicate the best and second-best DDI rates among *DDI-aware baselines*, respectively.

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prescriptions. We therefore follow prior work and evaluate DDI as an external safety metric, without explicitly optimizing for it.

Table 8 reports DDI rates as an external safety indicator. We highlight DDI-aware baselines separately since they explicitly optimize for drug–drug interaction reduction, while SpecTD-MR focuses on disease evolution modeling and does not incorporate DDI constraints. Notably, SpecTD-MR achieves competitive DDI rates despite not using any drug–drug interaction constraints, suggesting that modeling disease evolution implicitly reduces unsafe combinations. Incorporating DDI-aware regularization or context-dependent interaction modeling into our dependency framework is a promising direction for future work.

5.8 Limitations and Future Work

A key limitation lies in the interpretation of the learned dependency matrices. Although the proposed mechanism provides useful visual cues for inspecting model behavior, the dependency scores are computed from attention-based interactions and should not be regarded as faithful causal explanations. Attention weights may reflect predictive associations rather than true pathological mechanisms. Therefore, our interpretability analysis is intended as qualitative model inspection. Future work will incorporate more rigorous explanation methods, clinician-centered evaluation, and causal validation to better assess whether the inferred dependencies correspond to clinically meaningful disease progression.

Another limitation of the current population-level prior is that its reliability may vary across diagnosis frequencies. Although the population-level prior is helpful or neutral for low- and medium-frequency diagnoses, our frequency-stratified analysis shows that it may over-regularize high-frequency diagnosis patterns. This observation also provides motivation for future research along this direction.

6 Conclusion

In this paper, we presented SpecTD-MR, a novel framework designed to address the limitations of disease spectrum fragmentation and implicit dependency modeling in medication recommendation. By synergizing the Disease Spectrum-aware Hypergraph Learning module with the Time-evolving Dependency Modeling module, our approach learns spectrum-aware representations and models temporal predictive associations across patient visits. Extensive experiments on real-world datasets demonstrate that SpecTD-MR clearly outperforms state-of-the-art baselines. Furthermore, by visualizing model-inferred association patterns between historical and current clinical conditions, our framework provides qualitative insights into recommendation behavior. For future work, we plan to explore feasible pathways for combining dependency modeling approaches with safety considerations.

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